Expectation-Maximization Algorithms

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Computing Estimates in the Maximization Step of EM Algorithms

In the E-Step of EM algorithms, at $(t+1)$th iteration, we calculate

$$Q_i(z^{(i)}) := p(z^{(i)} \mid x^{(i)}; \theta^{(t)}). \tag{1}$$

In the M-Step of EM algorithms, at $(t+1)$th iteration, we execute

$$\theta^{(t+1)} := \arg\max_{\theta} \sum_{i=1}^{n} \sum_{z^{(i)}=1}^{K} Q_i(z^{(i)}) \log \frac{p\left(x^{(i)}, z^{(i)}; \theta\right)}{Q_i(z^{(i)})}. \tag{2}$$

Recall that,

$$z^{(i)} \sim \text{ Multinomial}(\pi) \; i.e. \; p\left(z^{(i)} = j\right) = \pi_j \geq 0, \sum_{j=1}^{K} \pi_j = 1,$$

$$x^{(i)} \mid z^{(i)} = j \sim \mathcal{N}(\mu_j, \Sigma_j),$$

$$\theta = \{\pi_1, \dots, \pi_K, \; \mu_1, \dots, \mu_K, \; \Sigma_1, \dots, \Sigma_K\}.$$

Let

$$\mathbb{Q}(\theta) = \sum_{i=1}^{n} \sum_{z^{(i)}=1}^{K} Q_i(z^{(i)}) \log \frac{p\left(x^{(i)}, z^{(i)}; \theta\right)}{Q_i(z^{(i)})}.$$

To get $\arg\max_\theta \mathbb{Q}(\theta)$, we differentiate $\mathbb{Q}(\theta)$ w.r.t. different components of θ, set the derivatives to 0 and solve the equations.

First we simplify $\mathbb{Q}(\theta)$ as follows.

$$\begin{aligned}
\mathbb{Q}(\theta) &= \sum_{i=1}^{n} \sum_{z^{(i)}=1}^{K} Q_i(z^{(i)}) \log \frac{p\left(x^{(i)}, z^{(i)}; \theta\right)}{Q_i(z^{(i)})} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{K} Q_i(z^{(i)} = j) \log \frac{p\left(z^{(i)} = j\right) \cdot p\left(x^{(i)} \mid z^{(i)} = j\right)}{Q_i(z^{(i)} = j)} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{K} w_j^{(i)} \log \frac{\pi_j \cdot \mathcal{N}(x^{(i)}; \mu_j, \Sigma_j)}{w_j^{(i)}}; \; w_j^{(i)} \doteq Q_i(z^{(i)} = j), \text{ say} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{K} w_j^{(i)} \log \pi_j \cdot \mathcal{N}(x^{(i)}; \mu_j, \Sigma_j), \text{ keeping relevant terms only} \\
&= \sum_{i=1}^{n} \sum_{j=1}^{K} w_j^{(i)} \log \pi_j + \sum_{i=1}^{n} \sum_{j=1}^{K} w_j^{(i)} \log \mathcal{N}(x^{(i)}; \mu_j, \Sigma_j)
\end{aligned}$$

$w_j^{(i)}$'s are set fixed in the E-Step; To maximize $\mathbb{Q}(\theta)$, we have to maximize $\log \pi_j$ and $\log \mathcal{N}(x^{(i)}; \mu_j, \Sigma_j)$ for each i, j.

Optimum π_j's are the ones that maximize $\sum_{i=1}^{n} \sum_{j=1}^{K} w_j^{(i)} \log \pi_j$ subject to the constraint $\sum_{j=1}^{K} \pi_j = 1$. So we use Lagrange Multipliers.

Define

$$\mathcal{L} = \sum_{i=1}^{n} \sum_{j=1}^{K} w_j^{(i)} \log \pi_j + \lambda \left(\sum_{j=1}^{K} \pi_j - 1 \right).$$

Then

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \pi_j} = 0 &\Rightarrow \sum_{i=1}^{n} w_j^{(i)} \cdot \frac{1}{\pi_j} + \lambda = 0 \\
&\Rightarrow \pi_j = -\frac{1}{\lambda} \sum_{i=1}^{n} w_j^{(i)}
\end{aligned}$$

Now

$$\begin{aligned}
\sum_{j=1}^{K} \pi_j = 1 &\Rightarrow \sum_{j=1}^{K} -\frac{1}{\lambda} \sum_{i=1}^{n} w_j^{(i)} = 1 \\
&\Rightarrow -\frac{1}{\lambda} \sum_{j=1}^{K} \sum_{i=1}^{n} w_j^{(i)} = 1 \\
&\Rightarrow -\frac{1}{\lambda} \sum_{i=1}^{n} \underbrace{\sum_{j=1}^{K} w_j^{(i)}}_{1} = 1 \\
&\Rightarrow -\frac{n}{\lambda} = 1 \\
&\Rightarrow \lambda = -n
\end{aligned}$$

$$\boxed{\therefore\ \pi_j = \frac{1}{n} \sum_{i=1}^{n} w_j^{(i)}.}$$

Optimum μ_j's are the ones that maximize

$$\begin{aligned}
&\sum_{i=1}^{n} \sum_{j=1}^{K} w_j^{(i)} \log \mathcal{N}(x^{(i)}; \mu_j, \Sigma_j) \\
&= \sum_{i=1}^{n} \sum_{j=1}^{K} w_j^{(i)} \log \left(\frac{1}{(2\pi)^{\frac{p}{2}} |\Sigma_j|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} \left(x^{(i)} - \mu_j \right)' \Sigma_j^{-1} \left(x^{(i)} - \mu_j \right) \right\} \right) \\
&= -\sum_{i=1}^{n} \sum_{j=1}^{K} w_j^{(i)} \left(x^{(i)} - \mu_j \right)' \Sigma_j^{-1} \left(x^{(i)} - \mu_j \right), \text{ keeping relevant terms only}
\end{aligned}$$

OR minimize $\sum\limits_{i=1}^{n}\sum\limits_{j=1}^{K} w_j^{(i)} \left(x^{(i)}-\mu_j\right)' \Sigma_j^{-1} \left(x^{(i)}-\mu_j\right)$.

$$\begin{aligned}
&\frac{\partial}{\partial \mu_j} \sum_{i=1}^{n}\sum_{j=1}^{K} w_j^{(i)} \left(x^{(i)}-\mu_j\right)' \Sigma_j^{-1} \left(x^{(i)}-\mu_j\right) = \mathbf{0} \\
&\Rightarrow \sum_{i=1}^{n} \frac{\partial}{\partial \mu_j} w_j^{(i)} \left(x^{(i)}-\mu_j\right)' \Sigma_j^{-1} \left(x^{(i)}-\mu_j\right) = \mathbf{0} \\
&\Rightarrow \sum_{i=1}^{n} \frac{\partial}{\partial \mu_j} w_j^{(i)} \left(x^{(i)}-\mu_j\right)' \left(\Sigma_j^{-1} x^{(i)} - \Sigma_j^{-1}\mu_j\right) = \mathbf{0} \\
&\Rightarrow \sum_{i=1}^{n} \frac{\partial}{\partial \mu_j} w_j^{(i)} \left(x^{(i)\prime}\Sigma_j^{-1} x^{(i)} - x^{(i)\prime}\Sigma_j^{-1}\mu_j - \mu_j'\Sigma_j^{-1}x^{(i)} + \mu_j'\Sigma_j^{-1}\mu_j\right) = \mathbf{0} \\
&\Rightarrow \sum_{i=1}^{n} \frac{\partial}{\partial \mu_j} w_j^{(i)} \left(-\Sigma_j^{-1}x^{(i)} - \Sigma_j^{-1}x^{(i)} + 2\Sigma_j^{-1}\mu_j\right) = \mathbf{0} \\
&\Rightarrow \sum_{i=1}^{n} w_j^{(i)} \left(-2\Sigma_j^{-1}\left(x^{(i)}-\mu_j\right)\right) = \mathbf{0} \\
&\Rightarrow \sum_{i=1}^{n} w_j^{(i)} \left(x^{(i)}-\mu_j\right) = \mathbf{0} \\
&\Rightarrow \sum_{i=1}^{n} w_j^{(i)} x^{(i)} - \mu_j \sum_{i=1}^{n} w_j^{(i)} = \mathbf{0}
\end{aligned}$$

$$\boxed{\therefore\ \mu_j = \frac{\sum\limits_{i=1}^{n} w_j^{(i)} x^{(i)}}{\sum\limits_{i=1}^{n} w_j^{(i)}}.}$$

Optimum Σ_j's are the ones that maximize

$$\begin{aligned}
&\sum_{i=1}^{n}\sum_{j=1}^{K} w_j^{(i)} \log \mathcal{N}(x^{(i)};\mu_j,\Sigma_j) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{K} w_j^{(i)} \log\left(\frac{1}{(2\pi)^{\frac{p}{2}}|\Sigma_j|^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}\left(x^{(i)}-\mu_j\right)' \Sigma_j^{-1}\left(x^{(i)}-\mu_j\right)\right\}\right) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{K} w_j^{(i)} \left(-\frac{p}{2}\log(2\pi) - \frac{1}{2}\log|\Sigma_j| - \frac{1}{2}\left(x^{(i)}-\mu_j\right)' \Sigma_j^{-1}\left(x^{(i)}-\mu_j\right)\right) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{K} w_j^{(i)} \left(-\log|\Sigma_j| - \left(x^{(i)}-\mu_j\right)' \Sigma_j^{-1}\left(x^{(i)}-\mu_j\right)\right), \text{ keeping relevant terms only} \\
&= \sum_{i=1}^{n}\sum_{j=1}^{K} w_j^{(i)} \left(\log|\Sigma_j^{-1}| - \left(x^{(i)}-\mu_j\right)' \Sigma_j^{-1}\left(x^{(i)}-\mu_j\right)\right)
\end{aligned}$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{K} w_j^{(i)} \ln\left(\sum_{\beta=1}^{p} \sigma_j^{\alpha\beta}\Sigma_j^{\alpha\beta}\right) - \sum_{i=1}^{n}\sum_{j=1}^{K} w_j^{(i)}\left(\sum_{\alpha=1}^{p}\sum_{\beta=1}^{p} \sigma_j^{\alpha\beta}\left(x_\alpha^{(i)} - \mu_{j\alpha}\right)\left(x_\beta^{(i)} - \mu_{j\beta}\right)\right) \tag{3}$$

where

$$\begin{aligned}
\Sigma_j &= ((\sigma_{j,\alpha\beta}))_{p\times p},\\
\Sigma_j^{-1} &= ((\sigma_j^{\alpha\beta}))_{p\times p},\\
|\Sigma_j^{-1}| &= \sum_{\beta=1}^{p} \sigma_j^{\alpha\beta}\Sigma_j^{\alpha\beta},\ \ \Sigma_j^{\alpha\beta} \text{ being the } (\alpha,\beta)\text{th cofactor of } \Sigma_j^{-1}.
\end{aligned}$$

Differentiating (3) w.r.t. $\sigma_j^{\alpha\beta}$ and setting that to 0, we have,

$$\begin{aligned}
&\sum_{i=1}^{n} w_j^{(i)} \frac{\Sigma_j^{\alpha\beta}}{\sum_{\beta=1}^{p} \sigma_j^{\alpha\beta}\Sigma_j^{\alpha\beta}} - \sum_{i=1}^{n} w_j^{(i)}\left(x_\alpha^{(i)} - \mu_{j\alpha}\right)\left(x_\beta^{(i)} - \mu_{j\beta}\right) = 0\\
\Rightarrow &\sum_{i=1}^{n} w_j^{(i)} \underbrace{\frac{\Sigma_j^{\alpha\beta}}{|\Sigma_j^{-1}|}}_{\Sigma_{j,\beta\alpha}} = \sum_{i=1}^{n} w_j^{(i)}\left(x_\alpha^{(i)} - \mu_{j\alpha}\right)\left(x_\beta^{(i)} - \mu_{j\beta}\right)\\
\Rightarrow &\Sigma_{j,\beta\alpha} = \frac{\sum_{i=1}^{n} w_j^{(i)}\left(x_\alpha^{(i)} - \mu_{j\alpha}\right)\left(x_\beta^{(i)} - \mu_{j\beta}\right)}{\sum_{i=1}^{n} w_j^{(i)}}
\end{aligned}$$

$$\boxed{\therefore\ \Sigma_j = \frac{\sum_{i=1}^{n} w_j^{(i)}\left(x^{(i)} - \mu_j\right)\left(x^{(i)} - \mu_j\right)'}{\sum_{i=1}^{n} w_j^{(i)}}.}$$